STAT 2102 · Practice Midterm 2: Food Delivery

Unofficial practice test in the style of Homework 2 · 50 points · suggested time 75 minutes

Instructions. Answer every question in one Quarto (.qmd) file and render it to PDF. Show the code and its output for each part, and answer the question **in words** below the output. Use `set.seed()` where a question tells you to. Use vectorized code or the apply family instead of loops where you can. The answer key is a separate file, so try everything first. This is **simulated** practice data, not the course's data set.

Problem 1: Functions (20 points)

Each answer needs only a few lines of code.

1.1 (5 pts)

Each order is late with probability `p_late`, independently of the others. Write a function named **orders_until_late** with one argument, `p_late`, that simulates orders one at a time until the first late order and returns how many orders that took. Run it once with `p_late = 0.2` after `set.seed(1)`, then use `replicate()` to estimate the average over 10,000 runs. What value would you expect?

1.2 (5 pts)

A delivery app charges a fee based on distance: $2.99 for trips of 3 km or less, $4.99 for more than 3 km up to 7 km, and $6.99 for longer trips. Write a function named **delivery_fee** that takes a vector of distances and returns the fee for each one, without using a loop. Test it with `c(1.5, 3, 3.01, 7, 12)`.

1.3 (5 pts)

Orders arrive in the Morning, Afternoon, Evening or Night with probabilities 0.20, 0.30, 0.35 and 0.15. Write a function named **shift_orders** with one argument, `n`, that simulates the time slot of `n` orders and returns a list with (a) a frequency table that shows **all four** time slots in that order, and (b) the name of the busiest time slot. Test it with `set.seed(2)` and `n = 15`.

1.4 (5 pts)

Write a function named **late_summary** with arguments `n`, `mean_time`, `sd_time` and `limit` (default 45). It should simulate `n` delivery times from a normal distribution, and return a named vector with the **proportion of deliveries longer than** `limit` and the **first quartile, median and third quartile** of the simulated times. Test it with `set.seed(3)`, `n = 200`, `mean_time = 38`, `sd_time = 9`. What is the exact probability that one delivery is late?

Problem 2: Food delivery data (30 points)

The file `food_delivery.csv` contains 500 orders from a food delivery app that wants to understand what makes deliveries slow. Some values are missing, as in real delivery data. The variables are:

Variable	Type	Description
`Order_ID`	Numeric	Unique identifier of the order
`Distance_km`	Numeric	Delivery distance (km)
`Weather`	Categorical	Clear, Foggy, Rainy, Snowy, Windy (some missing)
`Traffic_Level`	Categorical	Low, Medium, High (some missing)
`Time_of_Day`	Categorical	Morning, Afternoon, Evening, Night
`Vehicle_Type`	Categorical	Bike, Scooter, Car
`Preparation_Time_min`	Numeric	Time the restaurant took to prepare the order (minutes)
`Courier_Experience_yrs`	Numeric	Courier's experience (years; some missing)
`Delivery_Time_min`	Numeric	Total delivery time (minutes)

No file? Run the code in the Appendix: it creates exactly the same data and saves `food_delivery.csv` in your working directory.

2.1 (2 pts)

Import `food_delivery.csv` into R, converting all character variables into factors. Display the structure of the data, and report how many values are missing in each column.

2.2 (3 pts)

What is the average delivery time of all orders? How does the average delivery time compare across vehicle types? Additionally, among **Evening** orders only, how does the average delivery time compare across vehicle types?

2.3 (2 pts)

Which order took the longest to deliver (give its index)? Display that whole row. Conversely, which order was delivered the fastest?

2.4 (3 pts)

What is the median delivery time for each traffic level? Answer the same question again, now also taking Weather into account (every combination of traffic level and weather). How many orders are left out of this table, and why?

2.5 (2 pts)

What is the average courier experience? (Watch out for missing values.) What is the average courier experience for each vehicle type?

2.6 (4 pts)

Quartiles. Find the first, second and third quartiles of Distance_km. Compare the average delivery time of orders **longer than the third quartile** with that of orders **shorter than the first quartile**. Then add a new factor variable named `Distance_Group` with levels Short (at or below Q1), Medium (above Q1, up to Q3) and Long (above Q3), without modifying Distance_km, and report how many orders fall in each group.

2.7 (3 pts)

Compare delivery times across weather conditions using boxplots. Customize the boxplots by adding labels, changing the colors and adding notches. Are there any outliers?

2.8 (3 pts)

Provide an overview of traffic levels using a barplot. Display the bars in the order Low, Medium, High, make them horizontal, and customize the plot with labels and colors. Are the missing traffic values shown?

2.9 (3 pts)

Create side-by-side barplots showing the number of orders of each vehicle type for each time of day (in the order Morning, Afternoon, Evening, Night). Include a legend, and customize the plot as in the previous question.

2.10 (3 pts)

Plot delivery time against distance. Label the plot and axes properly. Use triangular points of size 0.9, colored by traffic level (three colors), add a legend and a regression line. Describe the relationship you see.

2.11 (2 pts)

Combine a histogram of delivery time (about 10 bins) and a boxplot of preparation time into a single image. Customize both plots as in Question 2.7 (no notch on the histogram).

Appendix: create the data set

Running this code once creates the same 500 orders every time (because of `set.seed`) and writes `food_delivery.csv`. Then start at Question 2.1.

```
set.seed(2102)
n <- 500
Distance_km  <- round(runif(n, 0.5, 20), 2)
Weather      <- sample(c("Clear", "Rainy", "Snowy", "Foggy", "Windy"), n, replace = TRUE,
                       prob = c(0.45, 0.20, 0.10, 0.10, 0.15))
Traffic_Level <- sample(c("Low", "Medium", "High"), n, replace = TRUE, prob = c(0.35, 0.40, 0.25))
Time_of_Day  <- sample(c("Morning", "Afternoon", "Evening", "Night"), n, replace = TRUE,
                       prob = c(0.20, 0.30, 0.35, 0.15))
Vehicle_Type <- sample(c("Bike", "Scooter", "Car"), n, replace = TRUE, prob = c(0.45, 0.30, 0.25))
Preparation_Time_min   <- sample(5:29, n, replace = TRUE)
Courier_Experience_yrs <- sample(0:9, n, replace = TRUE)
Delivery_Time_min <- pmax(10, round(10 + 2.8 * Distance_km + 0.9 * Preparation_Time_min +
                      6 * (Traffic_Level == "Medium") + 14 * (Traffic_Level == "High") +
                      5 * (Weather %in% c("Rainy", "Foggy")) + 9 * (Weather == "Snowy") -
                      0.6 * Courier_Experience_yrs + rnorm(n, 0, 8)))
# some values are missing, like in real delivery data
Weather[sample(n, 15)] <- NA
Traffic_Level[sample(n, 15)] <- NA
Courier_Experience_yrs[sample(n, 15)] <- NA
delivery <- data.frame(Order_ID = 1:n, Distance_km, Weather, Traffic_Level, Time_of_Day,
                       Vehicle_Type, Preparation_Time_min, Courier_Experience_yrs,
                       Delivery_Time_min)
write.csv(delivery, "food_delivery.csv", row.names = FALSE)
```